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Transparent Autonomy and Human Work Interaction Design

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Abstract. To acquire an appropriate level of trust between workers and between workers and autonomous systems, supporting transparency is desirable to convey awareness of each other's intents, reasoning processes, and future tasks. To achieve transparency, existing models recommend bi-directional communication or, at least, providing an explanation about the AI outputs and processes. In consequence, workers acquire new tasks to collaborate with non-human beings. This workshop proposes to investigate the impact of transparency mechanisms, which are desirable for efficient collaboration, on the work of people and their experience through the HWID lens.

Keywords: human work interaction design · transparency · teamwork · collaboration.

1 Introduction

Human Work Interaction Design (HWID) is concerned with designing usable and empowering systems, as well as positive experiences for workers by integrating HCI interaction design methods and work analysis. Contributions in this domain cover the design of automation and AI systems to support workers, as evidenced by dedicated conferences (e.g., HWID 2018: Designing Engaging Automation [2]; HWID 2021: AI and Designing for Positive Work Experiences in Low Desire Society [3]) and workshop publications (e.g., [9,10,11,5]). One challenge identified when designing such a system is enabling an appropriate level of trust to avoid under-reliance (e.g., workers reject a correct system's recommendation) or over-reliance (e.g., workers accept an incorrect system's classification) in automation [6]. To support effective collaboration with automation and AI systems,

one factor is promoting understanding of individuals' and teams' behavior and accuracy, a concept known as transparency [4,7]. Within this workshop, we propose to focus on the transparency of autonomous systems aimed at collaborating with workers as companions. Specifically, we seek to examine the experience, new tasks, or even new roles, undertaken by workers when working with transparent artificial teammates in various work sectors encompassing industry, research, education, and art.

2 Workshop Objectives

Within this workshop, we propose:

- To explore models and methods (including novel methods, e.g., art-based research) for human work interaction design of transparent autonomy, and approaches to adapt workers' skills and tasks to work within these teams,
- To collect case studies that investigate transparent human-autonomy teams through the HWID lens,
- To start a reflection on guidance for human work interaction design of transparent autonomy.

3 Workshop Topic Details

Transparency within a remote work setting where a team is collaborating through tools can be defined as processes of negotiations, interpretation, and consequence management involving communication of information (e.g, thinking process, tasks, concerns, state of the work) between team members for shared awareness of work progress, sharing, and feedback [12]. Within those settings, aspects concerning continuous visibility, visualization, and understanding of shared information, engagement, work ownership, surveillance of workers, and drawback effects (e.g., unsolicited interventions) composed the experience of transparency [12]. Different challenges arise when transparency concerns artificial intelligent systems added as team members (e.g, Human-AI Teaming). Indeed, transparency is defined differently when teams composed of humans collaborating with intelligent systems are studied and are no longer considered as tools but as teammates. Regarding decision-making tasks in mixed-initiative human-AI teaming, transparency is defined as 'bi-directional transparency'. In such team, transparency is supported by the bi-directional (i.e., from the system to the human and vice-versa) communication of the team decision-making processes (i.e., systems' intent, reasoning, future plans, uncertainties; humans' intents, constraints and objectives) and their shared tasks and interactions [4]. Lyons [7] defines this bi-directionality as 'robot-of-human' and 'robot-to-human' transparencies. Robot[7]-to-human transparency refers to the autonomous system communica-

[7] 'Robot' refers to an autonomous system (i.e., "a system that must operate autonomously in a dynamic environment under some conditions of uncertainty" [7])

tion of information for human understanding of the system (including an intentional model, task model, analytical and environmental models). Robot-of-human transparency refers to the communication of the autonomous system's understanding of its role and the human's role, as well as cognitive, emotional, and physical states (i.e., teamwork and a human state model). In the context of explainable AI algorithms, transparency can be defined as enabling users to see into black-boxes through design [8] by generating a "human-readable explanation for the model's decision" [1].

These new tasks, such as acquiring intents, understanding decision-making processes, and correcting the artificial teammate, impact workers' daily tasks and experience. These additional tasks and functions are described in transparency models. However, the impact of these changes on workers' tasks, goals, roles, and collaboration with other human teammates has not been investigated enough in the HWID community to allow alignment on guidance. When collaborating with an autonomous teammate, working with a transparent one is desirable; however, we believe that maintaining a sense of people's roles or creating new, desirable ones within this collaboration is also essential.

4 Expected Outcomes

To date, the HWID community has proposed a framework for collaboration with automation [10] and has investigated transparency experiences in remote human teamwork mediated by tools [12]. No models or guidance have been produced specifically addressing the issue of transforming workers' roles, tasks, and experience with transparent systems. We propose to initiate a reflection on the relevance and current state of research within the HWID field regarding transparent human work interaction designs with AI or autonomous systems. At the workshop, the organizers will invite the participants to continue their research activities with the aim of co-authoring a paper. The goal is to communicate the outcomes of this workshop to the broader HWID community, with a view to guiding this topic.

5 Target Audience

This workshop on transparent autonomy and Human Work Interaction Design invites HCI researchers and practitioners working on Human Work Interaction Design, particularly those focusing on designing positive experiences in human-autonomy collaboration in the workplace, to join. This workshop also aims to involve early-stage researchers and PhD students, who are encouraged to present their work-in-progress. To participate as a presenter, a 4-page position paper must be submitted and accepted.

6 Organizing Committee

The workshop is organized by IFIP TC13 WG 6 Human Work Interaction Design (HWID). The organizing committee is composed of:

Jose Abdelnour Nocera: Full Professor at the University of West London, UK. He is Head of The Sociotechnical Centre for Internationalisation and User Experience. He is also HWID vice-chair and co-chair of the sociotechnical specialist group for the British Computer Society. His research focuses in the sociotechnical and cultural aspects of systems design, development and use.

Elodie Bouzekri: Assistant Professor at the University of Brest, CNRS, Lab-STICC, France. Her research area, Human-Computer Interaction, focuses on the analysis and design of human-autonomy collaboration. She is Chair of the IFIP TC 13 Working Group 6 (WG 13.6) on Human Work Interaction Design (HWID).

Torkil Clemmensen: Professor at Copenhagen Business School, Denmark. PhD in human factors psychology. NordiCHI steering board member, senior editor for AIS Transactions of Human-Computer Interaction, HWID vice-chair and IFIP nat. repr. Contributes to Human-Computer Interaction, Design, Software Engineering, and Information Systems.

Arminda Guerra Lopes: Arminda Guerra Lopes is a professor at the Polytechnic Institute of Castelo Branco, Portugal. She has also held key leadership positions including vice director and president of both scientific and pedagogical boards. She earned her Ph.D. in Human-Computer Interaction from Leeds Metropolitan University in the United Kingdom. Arminda Guerra Lopes's research spans multiple interconnected areas within Human-Computer Interaction and Research Methodologies. It focuses on Social Informatics, Interaction Design, and quality-of-life technologies. She has contributed significantly to understanding Creativity and Innovation in technology contexts and has extensive experience in Collaborative Work research.

Antony William Joseph: Post Doctoral Researcher at Aalto University, Department of Design, School of Arts, Design and Architecture, Finland. His areas of interest inlcude Human-Centred Design, User Experience, Usability, Employee Well-being, and Meaningfulness of Work. Currently, his research focuses on developing new means to interact with automated system in the case of remote operation of autonomous ships and making the future of work with a more enriching user experience for the employees.

Adriana Moreno Rangel: Associate professor, researcher, and multimedia artist with academic and artistic work in the areas of art and technology, culture, and education. She holds a Ph.D. in Multimedia from the University of Lisbon, and her research involves collaborative design, visual programming, and the re-

lationship between art, technology, nature, and cognitive neuroscience. She is a member of research groups such as Artefacto (UFRR) and CIEBA (ULisboa). Her work includes the creation of interactive and installation-based products, oral narratives, and hybrid culture, as well as outreach projects with actions that promote the appreciation of migrant and refugee artists. She has experience in art-based research methodologies and creative processes that connect contemporary art, science, and technology.

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